

Amal V S

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EXPERIENCE

Metric Tree Labs

Kochi, Kerala

Sr. Software Development Engineer

Nov 2022 – Present

- Proficient in Node.js stacks (Express, NestJS, ReactJS) for backend development.
- Developed various stacks on Node.js, integrating data sources like CloudWatch, ELK, and databases such as PostgreSQL and MongoDB.
- Utilized various AWS resources such as Lambda, CloudWatch, to develop a dashboard displaying real time metrics.
- Experienced in designing and architecting high-scale social media and financial transaction applications.
- Skilled in system architecture with high-scale solutions using **Kafka** for event streaming and **Redis** for caching and real-time data processing.
- Built and managed CI/CD pipelines with **GitLab CI/CD** for automating Git workflows and deployment processes
- Experienced in maintaining multiple environments, ensuring smooth transitions and releases across development, staging, and production.
- Mentored junior developers, fostering growth and knowledge-sharing across backend and DevOps skills.
- Experienced in managing teams using **Agile** practices and **Jira** for project tracking, sprint planning, and task management, aligning project goals and priorities effectively.

SteamCube Pvt LTD

Kochi, Kerala

Jr. Software Development Engineer

Jan 2021– Nov 2022

- Leveraged Node.js, MongoDB, Kafka, and InfluxDB to develop a high-scale EdTech application that monitors and calculates study-level metrics for students.
- Architected and maintained multiple microservices, ensuring scalable and reliable system performance.
- Architected and maintained multiple microservices, orchestrating them within a Kubernetes (K8s) cluster for seamless scaling and resilience.
- Implemented the Factory Design Pattern to promote clean code and enhance scalability within the repository layer.

PROJECTS

Forum IAS

MTL, Kochi

Team Lead

Jun 2020 – Present

- Developed a highly scalable social media application like X app exclusively for IAS officers and IAS students.
- Designed a system for real-time updates on likes and comments using Socket.IO for live interactions and Kafka for efficient data streaming.
- Designed a system utilizing multiple databases for quick data access, with calculations sourced from a central "source-of-truth" (SOT) database to ensure consistency and accuracy.
- Architect a highly scalable chat module with all the core features of WhatsApp, utilizing Socket.IO for real-time communication, Kafka for distributed event streaming, and Memcache for efficient caching.
- Collaborated with team members to adhere to Agile methodologies, ensuring timely deliverables, efficient resource allocation, and effective communication.
- Coordinated sprint planning and daily stand-ups, utilizing JIRA for tracking progress and ensuring alignment within the team.

ECHO-Women

MTL, Kochi

Team Lead

Feb 2024 – Jul 2024

- Developed an e-commerce mobile application using React Native for the frontend, with a Node.js backend and PostgreSQL as the database.
- Lead an 8-member cross-functional team, including QA, to develop an e-commerce mobile application using React Native, Node.js, and PostgreSQL
- **Designed and implemented a robust stock management algorithm**, optimizing inventory tracking and product availability across multiple categories

- **Collaborated with team members** to adhere to Agile methodologies, ensuring timely deliverables, efficient resource allocation, and effective communication.
- **Managed timezone-specific functionalities** for a UK-based application, ensuring accurate scheduling, notifications, and data consistency across different regions.
- Integration Stripe for the Payment Intent integration.

Ecaps Edge

Senior Backend Engineer

MTL, Kochi

Nov 2022 – Jan2024

- Developed a highly scalable multi-tenant financial transaction application using NestJS, PostgreSQL, and Redis, incorporating multiple third-party banking API integrations.
- Designed a microservices architecture for the application, ensuring that each tenant has its own database. Implemented dynamic database connection switching for processing requests based on the tenant, enhancing data isolation and performance.
- Created a durable provider for managing and maintaining request switching and caching of Sequelize database connections, optimizing performance and resource utilization
- Deployed the application on AWS EC2 within an Auto Scaling Group to ensure high availability. Configured CloudWatch for monitoring and established a CI/CD pipeline using GitLab for automated deployment and continuous integration.
- Engaged with clients to discuss and gather requirements for new features and implementations, while also taking on the responsibility of managing the team.

StemClass

Jr. Full-stack developer

SteamCube, Kochi

Jan 2021 – Nov 2022

- Stemclass is an AI-powered web application designed to help schools guide students into Industry 4.0 technologies. It enables schools to track student and teacher progress using AI-driven insights and analytics.
- Implemented WebSocket-based real-time voice and text communication with a room-based architecture, enabling efficient group interactions and seamless user experience.
- Experienced in cloud paradigms and deployments, with hands-on expertise in DigitalOcean for scalable and efficient application hosting.

EDUCATION

Amrita School of Arts And Sciences

Master of Computer Application (MCA)

Kochi, Kerala

May 2016 - May 2021

- **Concentrations:** web application development and Intelligence and Modeling/Simulations
- **GPA:** 6.9
- **Related Coursework:** Data Structures & Algorithms, Objects & Design, Computer Organization & Programming, Combinatorics, Machine Learning, Artificial Intelligence, Object-Oriented Programming, Statistics & Applications

RESEARCH AND CERTIFICATION

International Conference on Ubiquitous Computing and Intelligent Systems

17, April 2021

Real-Time Emotion Recognition from Facial Expressions Using Convolutional Neural Networks with the FER2013 Dataset

Project Management Assessment Certificate

Awarded by LearnTube

Sep 2024

SKILLS

Programming: Typescript, JavaScript, python, Bash, SQL, Node.js, MongoDB, Postgres, sequelize, Prisma

Frameworks: Express.js, Nest.js, React.js

Tools: DataGrep, AWS, DigitalOcean, GitLab ci/cd, Agile, JIRA, ELK

Other: Kafka, Redis, Memcache, Docker, K8s, Nginx, Caddy server, Self Host servers